FET Biosensor Device Simulation with the DDM.SPC Solver

Theory, Physics and Implementation of Four ISFET/BioFET Models

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Abstract

This document gives the physical and mathematical background for four Python programs that model *ion-sensitive* and *biologically-sensitive field-effect transistors* (ISFET/BioFET) — the dominant electronic transducers for label-free disease detection — built on the two-dimensional drift-diffusion + Schrödinger–Poisson device solver **DDM.SPC**. The four programs form a fidelity ladder: (i) `biosensor_isfet.py`, a MOS-capacitor front-end; (ii) `biosensor_isfet_transfer.py`, a full-device drain-current transfer-curve model; (iii) `double_layer.py`, a self-consistent electrolyte site-binding + Gouy–Chapman–Stern double-layer module; and (iv) `biosensor_isfet_gdl.py`, which couples the electrolyte module to the device solver to produce a physically realistic, *sub-Nernstian* biosensor. Rather than repeat every topic for every file, we present the shared device physics once, the electrolyte physics once, and then a tabular comparison that maps each program onto the theory, together with the simulated results and their application to disease detection.

Contents

1 Introduction and background

1.1 Field-effect biosensors

An ISFET, introduced by Bergveld in 1970, is a MOSFET whose metal gate is replaced by an electrolyte contacted through a reference electrode and a chemically sensitive insulator surface [1, 2]. The quantity being sensed — proton activity (pH), a specific ion, or the charge of a captured biomolecule (DNA strand, antigen, enzyme product) — modifies the electrostatic potential at the insulator/electrolyte interface. That interfacial potential adds directly to the effective gate bias, shifting the transistor threshold voltage V_{th} and hence the drain current. Because the readout is a purely electrical, label-free threshold shift, field-effect biosensors underpin point-of-care diagnostics: pH and metabolic panels, DNA-FET pathogen detection, ImmunoFET antigen assays (troponin, C-reactive protein), and enzymatic glucose sensing [6, 7].

1.2 The DDM.SPC solver

DDM.SPC (*Drift-Diffusion Model + Schrödinger–Poisson Correction*) is a general-purpose 2-D, steady-state, finite-element semiconductor device solver. It solves the coupled Poisson and electron/hole continuity equations in the quasi-Fermi-potential formulation with a fully coupled Newton method and the exact analytic Jacobian, and supports heterojunctions, SRH/Auger/radiative recombination, Fermi–Dirac statistics, an insulated (MOS) gate referenced to a metal work function, Schottky contacts, and an optional self-consistent Schrödinger–Poisson quantum-confinement correction. The insulated-gate model is the key enabler here: the analyte-induced interfacial potential enters exactly as a shift of the gate work function, so the whole ISFET/BioFET family is representable with the existing solver.

1.3 Scope of the four programs

The four programs progress from an idealised demonstration to a physically realistic biosensor:

1. `biosensor_isfet.py` — the MOS-capacitor front-end. Establishes the transduction principle (analyte $\rightarrow$ gate work function $\rightarrow$ surface potential / threshold) with a robust, mesh-free structured solve.

2. `biosensor_isfet_transfer.py` — the complete transistor. Computes drain-current transfer curves $I_D(V_G)$ and extracts the three figures of merit (threshold shift, sub-threshold swing, sensitivity).

3. `double_layer.py` — the electrolyte interface physics (site-binding + Gouy–Chapman–Stern) that the device solver lacks; it produces the surface potential ψ_0 as a function of pH, ionic strength, and bound biomolecule charge.

4. `biosensor_isfet_gdl.py` — couples (iii) into the device solve so the analyte-to-gate coupling becomes realistically *sub-unity* (sub-Nernstian pH response and Debye-limited biomolecule detection).

2 Shared device physics

All four programs rest on the same semiconductor transport core; we state it once.

2.1 Drift-diffusion in quasi-Fermi variables

The primary nodal unknowns are the normalised electrostatic potential $u = \psi/V_T$ and the electron/hole quasi-Fermi potentials $v = \phi_n/V_T$, $w = \phi_p/V_T$, where $V_T = k_BT/q$ is the thermal voltage. Carrier densities follow from Boltzmann statistics,

$$n = n_i\, e^{u-v}, \qquad p = n_i\, e^{w-u}, \tag{1}$$

so the law of mass action $np = n_i^2$ holds at equilibrium ($v = w = 0$). With the net scaled doping $C = (N_D - N_A)/n_i$, the three coupled Galerkin weak forms solved by DDM.SPC are Poisson's equation and the two steady-state continuity equations,

$$\nabla\cdot(\varepsilon\nabla\psi) = -q\,(p - n + N_D - N_A), \tag{2}$$

$$\nabla\cdot \mathbf{J}_n = +q\,R, \qquad \mathbf{J}_n = q\mu_n n\nabla\phi_n, \tag{3}$$

$$\nabla\cdot \mathbf{J}_p = -q\,R, \qquad \mathbf{J}_p = q\mu_p p\nabla\phi_p, \tag{4}$$

where R is the net recombination rate (Shockley–Read–Hall + Auger; zero in the ideal-diffusion limit). Linear-triangle finite elements and 3-point Gauss quadrature assemble the residual and its analytic $3N \times 3N$ Jacobian, which is solved by damped Newton iteration.

2.2 The insulated MOS gate and the work-function knob

An insulated gate imposes a Dirichlet condition on the electrostatic potential only; the oxide carries no mobile charge. The gate node potential is set by the metal work function Φ_m and the applied voltage,

$$u_{\text{gate}} = \frac{V_{\text{gate}}}{V_T} + \frac{\Phi_{\text{ref}} - \Phi_m}{V_T}, \qquad \Phi_{\text{ref}} = \chi + \tfrac{1}{2}E_g + \tfrac{1}{2}V_T \ln\frac{N_c}{N_v}, \tag{5}$$

with Φ_{ref} the vacuum level at the reference intrinsic level. Equation (5) is the mathematical hinge of every model in this document: **a change $\Delta\Phi$ in the effective gate work function shifts the threshold voltage by the same amount**, $\Delta V_{\text{th}} = \Delta\Phi/q$ (to first order). In an ISFET the electrolyte supplies that shift through the interfacial surface potential ψ_0, so we identify

$$\Delta\Phi_{\text{gate}} = -\,q\,\psi_0. \tag{6}$$

2.3 Figures of merit

From a transfer curve $I_D(V_G)$ at fixed small drain bias, three figures of merit characterise a biosensor:

$$\text{threshold shift:}\quad \Delta V_{\text{th}}\ \text{(constant-current criterion)}, \tag{7}$$

$$\text{sub-threshold swing:}\quad S = \left(\frac{d\,(\log_{10} I_D)}{dV_G}\right)^{-1} \ \geq\ \ln(10)\,\frac{k_BT}{q} \approx 60\ \text{mV/dec}, \tag{8}$$

$$\text{sensitivity:}\quad \frac{\partial V_{\text{th}}}{\partial(\text{analyte})}. \tag{9}$$

The 60 mV/decade value in (8) is the room-temperature thermal limit; real short-channel devices exceed it.

3 Electrolyte interface physics (the G-DL model)

The bare device solver treats the gate potential as an ideal control; it has no electrolyte, so its analyte-to-threshold coupling is exactly unity. The module `double_layer.py` supplies the missing interfacial physics, which makes the coupling *sub-unity* — the behaviour real biosensors show.

3.1 Site-binding surface charge (2-pK model)

A functionalised oxide surface carries amphoteric hydroxyl sites that protonate or deprotonate depending on the local proton activity [3, 4, 5]. For total site density N_s with acid and base dissociation constants K_a, K_b, and surface proton activity $h_s = h_{\text{bulk}}\, e^{-q\psi_0/k_BT}$, the net surface charge is

$$\sigma_0(\psi_0) = qN_s \, \frac{h_s/K_a - K_b/h_s}{1 + h_s/K_a + K_b/h_s}. \tag{10}$$

The point of zero charge is $\text{pH}_{\text{pzc}} = \frac{1}{2}(\text{p}K_a + \text{p}K_b)$. The surface buffer capacity $\beta_{\text{int}} = -\, d\sigma_0/d\text{pH}_s$ (large when N_s is high and the pK spread small) sets how Nernstian the sensor is.

3.2 Gouy–Chapman–Stern double layer

The charged surface is balanced by the diffuse layer through a Stern (Helmholtz) capacitance C_{St} in series with the Gouy–Chapman diffuse charge,

$$\sigma_d(\psi_d) = -\sqrt{8\,\varepsilon_w\varepsilon_0 k_BT\, n_0}\; \sinh\!\Big(\frac{q\psi_d}{2k_BT}\Big), \qquad \psi_0 - \psi_d = \frac{\sigma_0}{C_{\text{St}}}, \tag{11}$$

where n_0 is the bulk ion number density. Charge neutrality $\sigma_0 = -\sigma_d$ closes the system; `double_layer.py` solves the scalar root

$$\sigma_0(\psi_0) + \sigma_d\big(\psi_0 - \sigma_0(\psi_0)/C_{\text{St}}\big) = 0 \tag{12}$$

self-consistently for ψ_0 at each pH (Brent's method; the site charge is bounded, so ψ_0 stays physical).

3.3 Sub-Nernstian pH response

The Nernst limit is $2.303\, k_BT/q \approx 59.5$ mV/pH at 298 K. The actual response is reduced by a dimensionless sensitivity parameter $\alpha \in (0, 1]$,

$$\frac{d\psi_0}{d\text{pH}} = -\,2.303\, \frac{k_BT}{q}\, \alpha, \qquad \alpha = \frac{1}{1 + \dfrac{2.303\, k_BT\, C_{\text{dl}}}{q^2\, \beta_{\text{int}}}}, \tag{13}$$

so a high buffer capacity β_{int} drives $\alpha \to 1$ (near-Nernstian, e.g. Ta_2O_5), while a low one gives a strongly sub-Nernstian surface (e.g. SiO_2).

3.4 Debye screening and the BioFET detection limit

A biomolecule of charge sheet density σ_{bio} captured a distance d from the surface is screened by the ionic double layer. In the linearised Debye–Hückel limit its contribution to the surface potential is

$$\Delta\psi_{\text{bio}} = \frac{\sigma_{\text{bio}}}{\varepsilon_w\varepsilon_0\, \kappa}\, e^{-\kappa d}, \qquad \lambda_D = \kappa^{-1} = \sqrt{\frac{\varepsilon_w\varepsilon_0 k_BT}{2q^2 N_A I}}, \tag{14}$$

with $\lambda_D \approx 0.30/\sqrt{I[\text{M}]}$ nm at room temperature. When the ionic strength I rises so that $\lambda_D < d$, the response collapses — the fundamental limitation of BioFET detection in physiological media.

4 Program comparison

Table 1 maps the four programs onto the physics of Sections 2–3.

Table 1: Tabular comparison of the four biosensor programs.

	biosensor_isfet	**biosensor_isfet_transfer**	**double_layer**	**biosensor_isfet_gdl**
Role	MOS-cap front-end	full device	electrolyte module	coupled full model
Structure	p-Si body + gate oxide	$n^+/p/n^+$ ISFET	— (interface only)	ISFET + electrolyte
Mesh	structured (built-in)	`mosfet_v3.msh` (gmsh)	none	`mosfet_v3.msh`
Core physics	Poisson + DD, MOS gate	+ continuity, $I_D(V_G)$	site-binding + GCS	device × electrolyte
Key equations	(2),(5)	(3)–(9)	(10)–(14)	(6),(13),(14)
Analyte model	$\Delta\Phi$ step	$\Delta\Phi$ series	$\psi_0(\mathrm{pH}, I, \sigma)$	$\psi_0 \to \Delta\Phi = -q\psi_0$
Output / FoM	ψ_s, ΔV_{th}	V_{th}, S, sensitivity	$d\psi_0/d\mathrm{pH}$, λ_D	sub-Nernstian $dV_{\mathrm{th}}/d\mathrm{pH}$
Coupling	unity (ideal)	unity (ideal)	sub-unity source	**sub-unity (realistic)**
Deps	DDM.SPC	DDM.SPC + gmsh	numpy/scipy	all of the above
Runtime	∼1 min	∼3 min (3 curves)	< 1 s	< 1 s; ∼7 min `-device`

5 Results

5.1 MOS-capacitor front-end

Figure 1 shows the silicon surface potential versus gate voltage for a clean surface and one carrying a +0.20 eV analyte-induced work-function step. The curve shifts rigidly by the same amount: the extracted threshold shift is $\Delta V_{\mathrm{th}} = +200$ mV, i.e. near-unity coupling, confirming Eqs. (5)–(6).

5.2 Full-device transfer curves

Figure 2 shows the $I_D(V_G)$ transfer curves (linear and semilog) for a series of analyte steps. Table 2 lists the extracted figures of merit. The coupling is exactly unity ($dV_{\mathrm{th}}/d\Phi = 1.000$), the ideal limit, because the bare device has no electrolyte; the sub-threshold swing $S \approx 135$ mV/dec exceeds the 60 mV/dec thermal limit, reflecting the short-channel geometry.

Table 2: Transfer-curve figures of merit ($W = 1$ μm, $V_D = 0.1$ V).

$\Delta\Phi$ (eV)	V_{th} (V)	ΔV_{th} (mV)	S (mV/dec)	I_{on} (A)
+0.00	−0.099	0	135	1.15×10^{-3}
+0.10	+0.001	100	135	1.09×10^{-3}
+0.20	+0.101	200	135	1.02×10^{-3}
Sensitivity $dV_{\mathrm{th}}/d\Phi = 1.000$ V/V (ideal, unity coupling).				

5.3 Electrolyte double layer and the realistic biosensor

Figure 3 shows the site-binding surface-potential response (left, its slope is the sub-Nernstian sensitivity) and the Debye-screening collapse of bound-charge detection with ionic strength (right). Table 3 summarises the sub-Nernstian pH sensitivity for four representative oxides, and Table 4 the screening collapse. When the electrolyte module drives the full device solve (`-device`), the

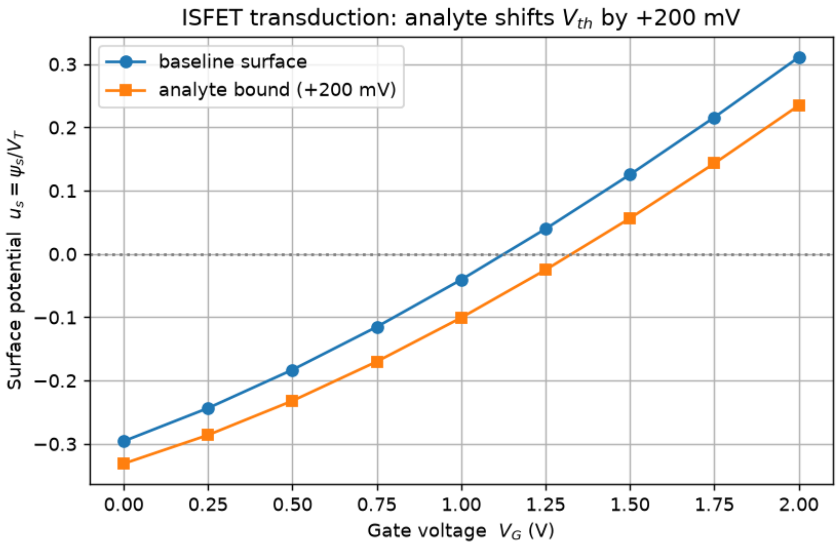

Figure 1: MOS-capacitor front-end (`biosensor_isfet.py`): analyte binding, modelled as a gate work-function step, rigidly shifts the surface-potential characteristic — the ISFET transduction principle.

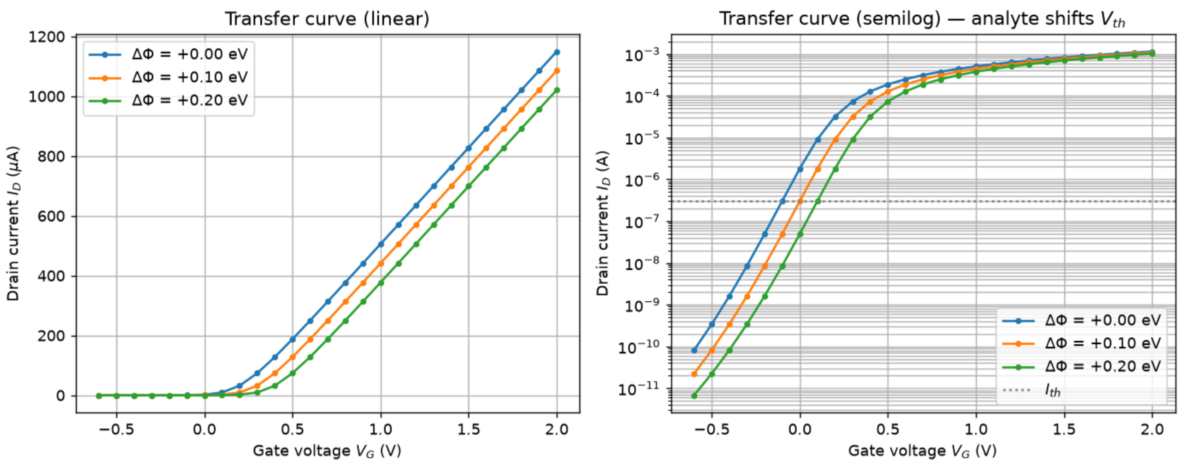

Figure 2: Full ISFET transfer curves (`biosensor_isfet_transfer.py`): analyte-induced work-function steps translate the $I_D(V_G)$ characteristic; the threshold shift is the sensor signal.

transistor threshold sensitivity is electrolyte-limited: $dV_{\mathrm{th}}/d\mathrm{pH} = 57.4$ mV/pH for Ta_2O_5 and 24.6 mV/pH for SiO_2 — both below the 59.5 mV/pH Nernst limit, unlike the ideal-coupling models.

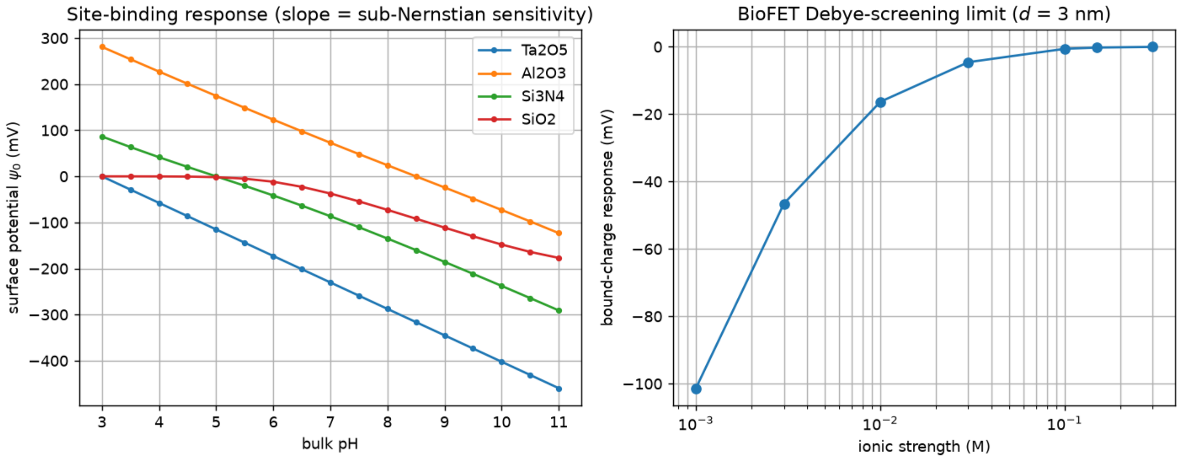

Figure 3: Electrolyte double-layer model (`double_layer.py` / `biosensor_isfet_gdl.py`). Left: site-binding surface potential $\psi_0(\mathrm{pH})$ — the slope is the sub-Nernstian sensitivity. Right: the Debye-screening limit — a bound charge at $d = 3$ nm becomes undetectable as the ionic strength shrinks λ_D below d.

Table 3: Sub-Nernstian pH sensitivity at pH 7, $I = 0.15$ M (Nernst limit 59.2 mV/pH).

oxide	mV/pH	α
Ta_2O_5	57.4	0.97
Al_2O_3	49.5	0.84
Si_3N_4	46.9	0.79
SiO_2	31.6	0.53

Table 4: BioFET Debye screening: bound charge -0.01 C/m^2 at $d = 3$ nm.

I (M)	λ_D (nm)	response (mV)
0.001	9.62	−101.3
0.010	3.04	−16.3
0.100	0.96	−0.61
0.150	0.79	−0.25

6 Applications to disease detection

The models map directly onto label-free diagnostic devices. In each case the analyte alters the interfacial surface potential ψ_0, which shifts the threshold via Eq. (6):

- **pH / metabolic panels** — direct proton response (Eq. (13)); oxide choice sets the sensitivity.
- **DNA-FET** — hybridisation adds phosphate-backbone charge; the Debye limit (Eq. (14)) dictates that low ionic strength or short probes are required.
- **ImmunoFET** — antigen–antibody binding charge; troponin/CRP cardiac and inflammatory markers.
- **Enzymatic (glucose) FET** — enzyme reaction changes local pH, read out as in the pH case.
- **Wide-bandgap / nanowire variants** — SiC/GaN surfaces for harsh-media and implantable sensing; thin-body nanowire BioFETs where the DDM.SPC Schrödinger–Poisson correction captures the confinement-enhanced sensitivity.

7 Limitations and outlook

The device solver is steady-state (no transient/AC, so binding kinetics and low-frequency noise are out of scope) and has no photogeneration or self-heating. The electrolyte module uses a mean-field 2-pK site-binding + Gouy–Chapman–Stern description; the per-oxide ($N_s, \mathrm{p}K_a, \mathrm{p}K_b$) parameters are illustrative values chosen to reproduce the known ISFET ordering and sensitivity ranges and should be re-fit to a given fabrication. Natural extensions are a transient continuity solve for kinetics/impedance and a quantum-corrected nanowire BioFET that exploits the solver's confinement model for enhanced sensitivity.

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